

THERMO 1

Thermo Fundamental Setup

Concept of spatiotemporal averaging: Consider length scales and time scales - when length scales/time scales are long enough, certain scales can be neglected/averaged (E.g. in a cup of water, you can ignore the water molecules on the surface (length) and molecular movement on the surface (time) and assume smoothness and constant height)

Concept of a system - Region of interest, separated by a boundary, real or imagined from the environment/surroundings

1. Open system: System that exchanges mass and energy with surroundings
2. Closed: No mass exchange, only energy
3. Isolated: No mass OR energy exchange with surrounds.

Composite systems can exist, composed of multiple system systems that have uniform properties. Furthermore, systems can contain restraints/constraints, which prevent certain thermodynamically favorable events from happening (e.g. clamps on a piston that keep it at a certain height, impermeable membrane, etc...).

Thermodynamics: The study of how a system changes over time and relative to other properties.

Zeroth law of thermodynamics: T is the quantity that is the driving force for heat flow

First: Conservation of energy ($\Delta U = Q + W$)

Second: Direction of time $\Delta S \geq 0$.

Third: Absolute 0, 0K, where all particle motion stops (remember, T is average KE)

Remember: $P = \frac{F}{A}$, and as such: $dW = Fdl = -PdV = \frac{mg}{A}(V_2 - V_1)$. $W > 0$ when done ON the system - expansion is work done BY the system. The third interpretation is for when blocks are placed on a piston.

Energy types: Kinetic (motion), Potential (position), Thermal (temperature), etc...

Intensive properties, unlike extensive, don't scale with quantity.

First Law

Energy is conserved such that $dE_{sys} + dE_{res} = 0$, so $Q_{sys} + W_{sys} = Q_{surr} + W_{surr}$.

Interaction between system and surrounding comes in the form of heat/work.

Heat capacity relates energy (U) to T, but is path dependent, unlike state functions, which are not ΔU , ΔS , etc. . . .

$$C_V = \left. \frac{\partial U}{\partial T} \right|_V$$
$$C_P = \left. \frac{\partial H}{\partial T} \right|_P = C_V + R$$

Processes occur because of driving forces, reversible processes have infinitesimal driving forces. Reversible processes can easily change direction with infinitesimal perturbations.

Shaft work - work on a closed system by turning an impeller (mechanical work)

Differential form of the first law: $dU = dQ + dW$, for reversible:

$dU = dQ - PdV = TdS - PdV$. If we account for chemical potential,

$dU = dQ - PdV + \sum_i \mu_i dN_i$. Other forms of work: BdI (magnetic), ψdq .

Total differential for U:

$$dU = \left(\frac{dU}{dT}\right)_V + \left(\frac{dU}{dV}\right)_T = C_V + \left(\frac{dU}{dV}\right)_V$$

Quantity enthalpy is useful for constant-pressure reversible processes. Definition is

$H = U + PV$, so then. $dH = dU + PdV + VdP$. For a constant pressure reversible process:

$dH = dU + PdV = dU - dW_{rev}$, $dH = dQ$.

State functions: ~~Path-dependent~~, the value only depends on the value of variables like T, P, composition. Examples are U, S, H, T, P. This can be taken advantage of by constructing simpler, ideal paths using isochoric, isobaric, adiabatic steps.

Phase Equilibrium Intro

Phases are continuous regions with homogeneous properties. Concept of chemical potential drives phase equilibrium, μ is the energy per unit molecule carried by a material that is transferring phases. At equilibrium, $\mu_1 = \mu_2$.

Gibbs phase rule: $F = 2 - \pi + N - R$, degrees of freedom depend on phases, components, reactions.

Saturated conditions are the conditions where 2 phases coexist (in variance analysis, $\pi = 2$)

Open systems:

Keep in mind our fundamental balance equation: $Acc = In - Out + Gen - Cons$.

Can relate mass flows to velocity and density: $\dot{m} = \rho u A = \frac{uA}{V_{specific}}$

Energy balance:

$$\frac{d(mU)_{cv}}{dt} = -\Delta\left[\left(U + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} + \dot{Q} + \dot{W}$$

English: Energy accumulation depends on the changes in internal, kinetic, and potential energy, as well as any heat/work input.

The work input is both PV work and shaft work ($\dot{W}_s$), which is from moving parts. Accounting for the PV term in the work, you can transform U to H to get the following:

$$\frac{d(mU)_{cv}}{dt} = -\Delta\left[\left(H + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} + \dot{Q} + \dot{W}_s$$

Certain simplifications:

$$\text{Steady-state : } 0 = -\Delta\left[\left(H + \frac{1}{2}u^2 + zg\right)\dot{m}\right]_{fs} + \dot{Q} + \dot{W}_s$$

$$\text{no external heat/work : } \Delta H = \dot{Q} + \dot{W}_s$$

Phase Equilibrium More Basics

Phase diagrams show the variation of phase with P, V, T (T variation on a PV diagram is shown via isotherms)

Principle of corresponding states: Use reduced units to compare substances: $T_r = \frac{T}{T_c}$, etc...

Ideal Gases - First Law Manipulations

Critical property in ideal gases (and something can be approximated) - $U^{ig} = U(T)^{ig}$.

For IG: $dU = C_v dT$, $dH = C_p dT$.

Since $PV = RT$ (when talking specific volume) $dW = -PdV = -RTdV$.

Also, $V = \frac{RT}{P}$, $dV = \frac{R}{P} dT - \frac{RT}{P^2} dP$

For IG only:

$$dU = dQ + dW, dQ = dU - dW = dU + PdV$$

$$dW = -PdV = P\left(\frac{R}{P} dT - \frac{RT}{P^2} dP\right) = RdT - \frac{RT}{P} dP.$$

$$dQ = dU - dW = +RdT - \frac{RT}{P} dP + C_v dT = (C_v + R)dT - \frac{RT}{P} dP = C_p dT - \frac{RT}{P} dP$$

Now consider a couple theoretical limits:

Isothermal = Constant T, so $dT = 0$. $dH = C_p dT = 0$, $dU = C_v dT = 0$, so $dH = dU = 0$ ($\Delta H = \Delta U = 0$).

$dU = dQ + dW = 0$, $dQ = -dW = PdV = RT \frac{dV}{V}$. (Can write these differentials in nondifferential form: $Q = -W = \int PdV$)

English: Constant temp means that there are no changes in internal energy or enthalpy, so Q and W can be directly related.

Isobaric = Constant P, $dP = 0$. $dW = -PdV$, or $W = -\int PdV = -P \int dV = -P\Delta V$

$dQ = dH = C_p dT$, $dU = C_v dT$.

English: At constant pressure, we can easily calculate enthalpy (C_p is a constant-pressure capacity), and we can know that $Q = \Delta H$. Work can be more easily calculable, using just the volume change.

Isochoric - Constant V, no work because $dV = 0$, so $PdV = 0$. $dU = dQ + \cancel{dW} = dQ = C_v dT$

English: At constant volume, there's no work (PdV work, ideal gas), so $dU = dQ$, both can be gotten from $C_v \Delta T$.

Adiabatic - No heat energy change ($dQ = 0$), so

$dU = dW$, $dU - dW = 0 = C_v dT + PdV = C_v dT + \frac{RT}{V} dV$. Rearrange: $C_v \frac{dT}{T} = -R \frac{dV}{V}$.

Solve:

$$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{R/C_v}$$

With $\gamma = \frac{C_p}{C_v}$, PV^γ would be constant.

English: So, in adiabatic situations, $Q = 0$, which allows us to directly relate U and W, and put temperatures, pressures, and volumes in relation with one another

Non-Ideal EOS

Equations exist to model property profiles in non-ideal gases

Remember the compression factor: $Z = \frac{PV}{RT}$, it is 1 for ideal gases, but not for real ones due to intermolecular interactions

Virial equation of state: (Used to describe dilute gas behavior)

$$Z = B'P + C'P^2 + D'P^3, \text{ where } B' = \frac{B}{RT}, C' = \frac{C - B^2}{(RT)^2}, \text{ etc.} \dots$$

$$Z = 1 + \frac{B}{V} + \frac{C}{V^2}, \text{ etc.} \dots$$

Number of term determines accuracy.

Van der waals EOS: (Accounts for intermolecular forces)

$$P = \frac{RT}{V - b} - \frac{a}{V^2}$$

Liquids

Liquids: V can be thought of a function of P and T: $V = V(P, T)$

$$dV = \left(\frac{dV}{dT}\right)_P dT + \left(\frac{dV}{dP}\right)_T dP$$

$$\frac{dV}{V} = \beta dT - \kappa dP$$

$\beta = \frac{1}{V} \left(\frac{dV}{dT}\right)_P$, called volume expansivity, and $\kappa = -\frac{1}{V} \left(\frac{dV}{dP}\right)_T$, called isothermal compressibility.

β , κ are small for condensed phases, and 0 for incompressible fluids (constant V)

Deeper Look At U, H, Q

Different types of heat changes:

1. Latent heat: Heat of phase change
2. Heat of reaction: Heat evolution when species change/react
3. Sensible heat: Heat and cooling in response to non-phase changes w/o reaction

We've only thought of U and H as functions of T, but the total differentials would look like:

$$dU = \left(\frac{dU}{dT}\right)_V dT + \left(\frac{dU}{dP}\right)_T dP$$

$$dH = \left(\frac{dH}{dT}\right)_V dT + \left(\frac{dH}{dP}\right)_T dP$$

Of course for an ideal gas, the dP terms are neglected, so $dH = C_p dT$, $dU = C_v dT$, but remember that C_v , C_p , can also be functions of T, such that $\Delta H = \int C_p(T) dT$.

Latent heat effects: No temperature change.

Clausius-Clapeyron relates vapor-liquid phase change and the relevant properties:

$$\Delta H = T \Delta V \frac{dP^*}{dT}$$

The $\Delta V = V^V - V^L$, the molar volume change by phase change alone.

For a chemical change: $\Delta H = Q$.

In a reaction $\Delta H^o = \sum_i \nu_i \Delta H_{f,i}$, the enthalpy change of a reaction depends on the reaction stoichiometries and the heats of formation of the species

Second Law

Remember that

1. At nonstandard temps, the heat of formation then also depends on $C_p dT$ for the temp change
2. H_f for a naturally occurring element, H_2 gas, for instance, is 0.

When charting a process for a reaction at a certain T, you have to change the reactant and product heats of formation to the new temperature based on the C_p , and then subtract.

Remember the 2nd law: $\Delta S_{universe} \geq 0$ (only 0 for reversible (isentropic) processes. Statical mechanical interpretation of entropy $S = k_B \ln(\Omega)$ where μ is the number of microstates.

Entropy helps us understand reversibility - reversible processes have no entropy generation, irreversible processes do (thought experiment: why does reversing an irreversible process not make sense in terms of entropy? - It's because its reverse has a negative entropy change, which is not possible due to the second law)

$$dS = \frac{dQ_{rev}}{dT}, \text{ so } \Delta S = \int \frac{dQ_{rev}}{dT}$$

$$H = U + PV, dH = dU + d(PV) = dU + PdV + VdP$$

At constant pressure: $dH = dU + PdV = dU - dW = dQ_{rev} = TdS$.

$$\text{So, } TdS = dH = \left(\frac{dH}{dT}\right)_P + \left(\frac{dH}{dP}\right)_T dP = C_P dT.$$

$$\text{So, } dS = \frac{C_p}{T} dT, \left(\frac{dS}{dT}\right)_P = \frac{C_p}{T}.$$

At constant volume: $dW = 0$, $dU = dQ_{rev} = TdS$, so $TdS = C_v dT$, $\left(\frac{dS}{dT}\right)_V = \frac{C_v}{T}$

You can calculate changes in S similar to U and H, as it is a state function that has no path dependence. The above manipulations are exactly that, calculating an irreversible process using fictitious reversible processes.

So, $dS = \frac{dQ}{T} + dS_{gen}$, where $dS_{gen} = 0$ for reversible processes. For open systems, you add flow stream terms.

So remember, $dU = dQ_{rev} + dW = dQ_{rev} - PdV$, and $dH = dU - \cancel{PdV} + \cancel{PdV} + VdP$

From here:

$$dQ_{rev} = TdS = dH - VdP, dS = \frac{dH}{T} - \frac{V}{T}dP$$

For ideal gases, $dH = C_p dT$, $\frac{V}{T} = \frac{R}{P}$, so, $dS = \frac{C_p}{T} dT - \frac{R}{P} dP$.

$$\frac{\Delta S^{ig}}{R} = \int_{T_0}^T \frac{C_p}{RT} dT - \ln\left(\frac{P}{P_0}\right)$$

So, for isobaric processes ($dP = 0$):

$$\Delta S = \int_{T_0}^T \frac{C_p}{T} dT$$

And for isothermal processes ($dT = 0$):

$$\Delta S = nR \ln\left(\frac{V_f}{V_i}\right) = -nR \ln\left(\frac{P_f}{P_i}\right)$$

$$Q = -W \text{ (} dU = 0 \text{)} = \int TdS = T \int dS = nRT \ln\left(\frac{V_f}{V_i}\right) = -nRT \ln\left(\frac{P_f}{P_i}\right)$$

Irreversible processes result in increase in the universe's entropy. Isentropic processes are adiabatic and reversible.

In irreversible processes, there is lost work, work that could have been performed in reversible.

$$W_{lost} = W_{rev} - W_{actual}$$

Irreversibility comes from driving forces like gradients in P, T, C.

Closer look at second law:

Properties of entropy:

1. State function $S(E, \vec{X})$, where X is a set of extensive properties. Also, there is existence of some function $E(S, \vec{X})$ that shares similar functional properties.
2. Monotonically increasing function of E (U). (Existence of some function S(E,))
3. Differentiable and continuous as a function of E
4. Extensive and first-order homogeneous: $\lambda S(E, \vec{X}) = S(\lambda E, \lambda \vec{X})$.
5. If some state B is adiabatically accessible from state A, $S_B \geq S_A$ (just another way of describing that isentropic processes are adiabatic and all others have positive entropy changes)

Revisit reversibility - we think of the beginning and end, but let's dive further into the PATH they take

1. Reversible processes: Take a path involving a manifold of equilibrium states
2. Irreversible processes: Take a path involving a manifold of nonequilibrium states

Two key properties, then, that the second law dictates: (and think of the constraint as really anything creating irreversibility)

1. $S(E, \vec{X}, constraint) < S(E, \vec{X})$: Principle of entropy maximization
2. $E(S, \vec{X}, constraint) < E(S, \vec{X})$: Principle of energy minimization

Thermo calculus:

Total differentials allow us to understand the variations of different parameters:

$$dU = TdS - PdV \text{ is the most fundamental of these}$$

Transforms have been invented however, to put parameters in terms of more easily measurable items. These are called Legendre transforms, and you can modify them based on which variables you're controlling.

For instance, if you're controlling T, this is the Helmholtz free energy Legendre transform:

$$A = U - TS$$

Controlling for T and P? That's Gibbs: $G = U - TS + PV$

Let's work out their total differentials:

$$\begin{aligned} dA &= dU - d(TS) = TdS - PdV - TdS - SdT = -PdV - SdT \\ dG &= dU - d(TS) + d(PV) = TdS - PdV - TdS - SdT + PdV + VdP = -SdT + VdP \end{aligned}$$

Add in the other two parameters to get the most common differentials we have:

$$dU = TdS - PdV (+\sum_i \mu_i dN_i \text{ if we factor in chemical potential})$$

$$H = U + PV, dH = TdS - PdV + PdV + VdP = TdS + VdP$$

Using mixed derivatives, you can derive maxwell relations that relate hard-to-measure derivatives in terms of more palatable ones. You can use the total differentials for this. Derive them yourself before looking at the following.

$$\left(\frac{dP}{dT}\right)_V = \left(\frac{dS}{dV}\right)_T$$

$$\left(\frac{dS}{dP}\right)_T = -\left(\frac{dV}{dT}\right)_P$$

$$\left(\frac{dT}{dV}\right)_S = -\left(\frac{dP}{dS}\right)_V$$

$$\left(\frac{dT}{dP}\right)_S = \left(\frac{dV}{dS}\right)_P$$

Revisiting the total differential for S(T,P):

$$dS = \left(\frac{dS}{dT}\right)_P dT + \left(\frac{dS}{dP}\right)_T dP$$

Using the legendre transform for H, at constant P, $dH = TdS$, $\left(\frac{dH}{dT}\right)_P = C_p = T\left(\frac{dS}{dT}\right)_P$, so $\left(\frac{dS}{dT}\right)_P = \frac{C_p}{T}$

Then combining with the maxwell relation:

$$dS = \frac{C_p}{T} dT - \left(\frac{dV}{dT}\right)_P dP$$

Multiplicative rule (also known as triple product rule):

$$\left(\frac{dx}{dy}\right)_z \left(\frac{dy}{dz}\right)_x \left(\frac{dz}{dx}\right)_y = -1$$

$$\left(\frac{dV}{dT}\right)_P \left(\frac{dT}{dP}\right)_V \left(\frac{dP}{dV}\right)_T = -1$$

Can rearrange:

$$\left(\frac{dx}{dy}\right)_z = -\left(\frac{dz}{dy}\right)_x \left(\frac{dx}{dz}\right)_y$$

Useful when you know some of these derivatives (e.g. you know that $C_p = \left(\frac{dH}{dT}\right)_P$)

Residual properties:

Show the deviation from ideality - from some property M, $M^R = M - M^{ig}$.

Working this out for volume:

$$V^R = V - V^{ig} = V - \frac{RT}{P} = \frac{ZRT}{P} - \frac{RT}{P} = (Z - 1) \frac{RT}{P}$$

Often, EOS's can be used for residual properties.

THERMO 2

Entropy balance can be done on flow systems:

$$\Delta(\dot{m}S)_{fs} + \frac{d(mS)_{cv}}{dt} + \frac{dS_{surr}^t}{dt} = \dot{S}_G \geq 0$$

English: Total rate of entropy generation = Change of entropy in flow streams, Time rate of change of entropy in control volume + Time rate of change in the surroundings.

Deviations from ideality:

Repulsive forces assist expansion, attractive forces assist compression, at far distances, both forces die down. At moderate to high pressures, these changes can occur too (moderate = attractive forces dominate, high pressure = repulsive)

Equilibria (Phase), Chemical Potential

We looked at this in terms of the U total differential, but let's look at it again in terms of G, which is a better legendre transform as far as characterizing equilibria is concerned. Construct from scratch, keeping in mind that we're controlling T and P here.

$$G = U - TS + PV$$

$$dG = (TdS - PdV + \sum_i \mu_i dN_i) - TdS - SdT + PdV + VdP = VdP - SdT + \sum_i \mu_i dN_i$$

At equilibrium, and constant T, and P, $dG = 0$. This ends up yielding the result that at equilibrium of two phases a and b, $\mu_a = \mu_b$. μ , in other words, is molar Gibbs free energy.

Remember the legendre transform for Gibbs free energy: $dG = VdP - SdT$. So, making this molar, where μ is molar gibbs: $d\mu = V_m dP - S_m dT$

So, $(\frac{d\mu}{dP})_T = V_m$, $(\frac{d\mu}{dT})_P = -S_m$. Keep in mind that $S_{gas} > S_{liquid} > S_{solid}$.

If $dG < 0$, $d\mu < 0$ and the process is spontaneous. If $dG > 0$, $d\mu > 0$ and the process is spontaneous in the opposite direction.

Clapeyron equation relates molar properties to the pressure/temperature phase boundary:

$$\frac{dP}{dT} = \frac{\Delta S_m}{\Delta V_m}$$

For SLE: $\Delta S_{m,melt} = \frac{\Delta H_{m,melt}}{T_{melt}}$, so $\frac{dP}{dT} = \frac{\Delta H_{m,melt}}{T_{melt} \Delta V_{m,melt}}$

For VLE: $V_m = \frac{RT}{P}$, $\frac{d \ln P}{dT} = \frac{\Delta H_{m,vap}}{RT^2}$

For SVE: $V_m = \frac{RT}{P}$, $\frac{d \ln P}{dT} = \frac{\Delta H_{m,sub}}{RT^2}$

Phase rule: $F = C - P + 2 - R$, in english: Variance = components - phases + 2 - reactions

Txy/Pxy diagrams relate temperature and pressure in phase equilibrium - Tie lines can be used to relate isothermal and isobaric phase changes to composition. This is aided by the lever rule:

$$\frac{n_\alpha}{n_\beta} = \frac{d_\beta}{d_\alpha}$$

The ratio of the amounts of phase a and b can be found on a Txy/Pxy diagram by looking at how close/far away you are from each line, the *dew line for saturated vapor*, the *bubble line for saturated liquid*. So the mid point of the bubble due would actually be the mid point of 50% vapor, 50% liquid.

Normal Txy/Pxy diagrams never intersect, but there are exceptions called azeotropes.

1. At an azeotrope, $x = y = Z$ (Z is overall composition)
2. At an azeotrope, the slopes of the bubble/dew lines switch signs
3. At an azeotrope, the boiling points are the same of both components - cannot be separated

1. How to avoid an azeotrope? (A: Manipulate T and P before hitting it to AVOID)
4. Two types of azeotropes:
 1. Minimum boiling azeotrope: Unfavorable interactions between A and B lead to a positive deviation from Raoult's law that DESTABILIZE the liquid. Maximum in the Txy diagram ($\gamma > 1$)
 2. Maximum boiling azeotropy: Favorable interactions that stabilize due to a negative deviation from Raoult's law. ($\gamma < 1$)

Partial Molar Properties + Gibbs-Duhem

Partial molar properties are per-mole, per-species properties.

So for some property M, $M_i = M_i = \left[\left(\frac{dM}{dn_i} \right)_{P,T,n_i} \right]_{n_i}$.

So in the case of volume, for instance:

$$dV = V_A dn_A + V_B dn_B = \left(\frac{\partial V}{\partial n_A} \right) dn_A + \left(\frac{\partial V}{\partial n_B} \right) dn_B$$

$$V = V_A n_A + V_B n_B$$

Gibbs-Duhem eqn (M is a partial molar property):

$$\left(\frac{\partial M}{\partial P} \right)_{T,X} dP + \left(\frac{\partial M}{\partial T} \right)_{P,X} dT - \sum_i X_i dM_i = 0$$

Another way to look at a Gibbs-Duhem equation - we've been formulating things in terms of intensive*dExtensive (e.g. PdV, TdS)

$$0 = \sum \text{extensive} * d\text{Intensive} = -SdT + VdP - \sum_i N_i d\mu_i$$

How to derive - Write the euler relation for U, ($U = TS - PV + \dots$), derive, and then take out the terms from the first law differential. (So the opposite of the derivatives in the first law is, in essence, what you get)

Other forms of work:

1. Magnetic BdI .
2. Electrical ϕdq
3. Mechanical FdL (elongation/compression can be an application here)

Mixture Properties

Take some property M in a mixture - you can relate full and partial molar properties as follows in a mixture:

Take the examples of entropy and enthalpy

$$M^{ig} = \sum_i M_i^{ig} y_i$$

$$e.g. H^{ig} = \sum_i H_i^{ig} y_i$$

$$S_i = -R \ln P, dS_i = -R d \ln P$$

$$y_i = \frac{P_i}{P}$$

$$S = \sum_i y_i S_i - R \sum_i y_i \ln y_i$$

In ideal gas mixing processes:

1. $\Delta G < 0$
2. $\Delta S > 0$
3. $\Delta H = 0$

FUGACITY and Raoult's Law

Fugacity helps us understand deviations from ideality in terms of pressures. You can think of it as an "effective pressure"

Use the concept of residual properties to derive:

$$G^R = G - G^{ig} = RT \ln(f_i) - RT \ln(P) = RT \ln\left(\frac{f_i}{P}\right) = RT \ln(\phi)$$

ϕ is the fugacity coefficient:

1. At moderate pressure: $\phi < 1$, $f_i < P$, attractive forces reduce effective pressure
2. High pressures: $\phi > 1$, $f_i > P$, repulsive forces increase effective pressure

We've defined phase equilibrium in terms of chemical potential, but we can also say that define in terms of fugacity:

$$\begin{aligned} f_i^V &= f_i^L = f_i^* \\ \phi_i^V &= \phi_i^L = \phi_i^* \end{aligned}$$

ϕ_i^* , f_i^* correspond to pure vapor or liquid. Phases are in equilibrium when they have the same T, P, f.

An analog to the residual property is the excess property for solutions: $M^E = M - M^{id}$, the difference between a property and the ideal solution's property. When $M^E = 0$, $M = M^{id}$, and $\gamma \rightarrow 1$.

1. At moderate pressure: $\gamma < 1$, $f_i < P$, attractive forces reduce effective pressure
2. High pressures: $\gamma > 1$, $f_i > P$, repulsive forces increase effective pressure

Fugacities can be present in gas and liquid phases:

Gas phase:

$$G^R = G_i - G_i^{ig} = RT \ln\left(\frac{\hat{f}_i}{f_i^{ig}}\right)$$

Solution:

$$G^E = G_i - G_i^{id} = RT \ln\left(\frac{\hat{f}_i}{\hat{f}_i^{id}}\right)$$

We can define fugacity (mixture) and activity (solution) coefficients for non-pure species that partial molar properties/molar fractions:

$$\hat{\phi}_i = \frac{\hat{f}_i}{y_i P}, \quad \gamma_i = \frac{\hat{f}_i}{X_i f_i}$$

Partial fugacities can then be related to these coefficients, and residual free energy/excess free energy can be reformulated for non-pure components.

$$\begin{aligned}\hat{f}_i &= y_i P \hat{\phi}_i \\ \hat{f}_i &= X_i f_i \gamma_i \\ G_i^R &= RT \ln(\hat{\phi}_i), \quad G_i^E = RT \ln(\gamma_i)\end{aligned}$$

With a framework for evaluating excess and residual properties in both pure component and multi-component systems, we can now re-re-revisit phase equilibrium. Take VLE:

$$\begin{aligned}f^L &= f^V \\ y_i P \hat{\phi}_i &= \gamma_i X_i P_i^*\end{aligned}$$

P^* , as we remember, can be calculated via Antoine's eqn: $\ln(P^*) = A - \frac{B}{T-C}$.

We can get Raoult's law to relate liquid and vapor pressures at multiple levels of abstraction:

1. Ideal solution + ideal gas mixture ($\phi, \gamma \rightarrow 1$): $P_i = y_i P = X_i P_i^*$ Here, pressure = fugacity because we're in an ideal situation
 1. Remember that the $P_i = y_i P$ is Dalton's law of partial pressures
2. Ideal gas, real solution: $y_i P = \gamma_i x_i P_i^*$. (Called modified Raoult's law)
3. Ideal solution, real gas: $\Phi_i y_i P = x_i P_i^*$
4. Real solution real gas: $y_i P \Phi_i = \gamma_i X_i P_i^*$

Here, $\Phi_i \equiv \frac{\hat{\phi}_i}{\phi_i^* \exp[(RT)^{-1}(V_i'(P-P_i^*))]}$. As $P \rightarrow P^*$, $\Phi \rightarrow \frac{\hat{\phi}_i}{\phi_i^*}$

All are ways of wording the following equation which would be true for all situations:

$$\hat{\phi}_i y_i P = \gamma_i X_i f_i^L.$$

Remember Henry's law? $P_i = y_i P = H_i x_i$? When is it better? (A: very dilute solutions). Raoult's law is best for ideal gas/solution, but ofc nonidealities can be made up for with modified Raoult's laws.

Thermochemistry:

ΔH_{rxn} , or enthalpy of reaction depends on temperature, and state. Can be related to standard enthalpy of reaction by factoring heating/phase change enthalpies (for heating, of course, it would be $\int C_p dT$)

ΔH_f is the enthalpy change when a compound is formed. It is 0 for pure substances at their standard state.

Remember: $\Delta H_{rxn} = \sum_{products} \nu_i H_{fi} - \sum_{reactants} \nu_i H_{fi}$.

Phase change heats exist too: ΔH_f for fusion, or melting, ΔH_{vap} for vaporization, ΔH_{sub} for sublimation, ΔH_{soln} for dissolving a substance.

Equilibrium constant can be summarized in terms of fugacity and then into the form we know:

Take the reaction $aA + bB \rightarrow cC + dD$

First, we consider that at equilibrium, it always comes back to chemical potential: $\sum_i \nu_i \mu_i = 0$

$$\begin{aligned}K &= \Pi_i \left(\frac{\hat{f}_i}{f_i^0} \right)^{\nu_i} = \frac{[C]^c [D]^d}{[A]^a [B]^b} \\ \Delta G &= -RT \ln(K)\end{aligned}$$

Spontaneous process (review): $\Delta G < 0$, $\Delta \mu < 0$, $K > 1$.

Hess's law: Add+multiply reaction enthalpies, entropies, etc... (state functions) of multiple reactions that can sum up to get a target reaction to find the overall values. For equilibrium constants, you multiply them.